

Branch-Preserving Temporal Travel (BPTT)

One-Page Technical Summary • Conditional Mathematical / Phenomenological Framework

1. Research Question

BPTT asks a conditional question: if a physical mechanism capable of backward temporal displacement existed, could a traveller move from source history A at departure time t_d into a causally distinct history B at earlier time t_a , while preserving the pre-departure causal history of A?

2. Core Structure

$$A(t_d) \rightarrow B(t_a), t_a < t_d, A \neq B$$

A hypothetical quantum operation $\blacksquare_{A \rightarrow B}$ represents the branch transition. For an ordinary open-system operation, it is required to be completely positive and trace preserving (CPTP).

3. Central Mathematical Condition

Let $\blacksquare_A^{<t_d}$ denote the algebra of observables associated with the pre-departure causal history of A. BPTT imposes:

$$\blacksquare_{A \rightarrow B}^*(O_A) = O_A, \forall O_A \in \blacksquare_A^{<t_d}$$

By trace duality, $\text{Tr}[O_A \rho] = \text{Tr}[O_A \rho]$. Thus protected pre-departure observables retain their expectation values after the transition.

4. Minimal Quantum-Information Check

A three-register toy model (traveller T, source A, destination B) uses an ordinary SWAP between T and B. The traveller state transfers to B while A is unchanged. For an A-local observable, $[U_{TB}, O_A] = 0$, so its expectation value is preserved. This establishes mathematical compatibility of source-subsystem invariance with ordinary unitary state transfer.

Limitation: the SWAP model does not demonstrate time travel or physical branch transfer; A and B are abstract registers.

5. Relation to Existing Work

Framework	Relation to BPTT
Lewis / alternate histories	Alternate-history resolution is an established predecessor.
Deutsch D-CTC	Uses a density-matrix fixed-point consistency condition.
Branching spacetime	Already studies diverging causal histories.
Shoshany–Stober E-CTC	Closest modern comparison; uses unitary evolution and timeline-correlation structure with emergent decohered timelines.

6. Current Scientific Status

BPTT does **not** claim that backward time travel is physically realizable, that branch-to-branch travel has been observed, or that BPTT is an established new physical theory. The present contribution is a formal language for source-history preservation and for separating ordinary quantum state transfer from the stronger, currently unestablished notion of controllable transfer between decohered histories.

7. Open Problems / Required Tests

Requirement	Current status
Physical branch-transition mechanism $\blacksquare_{A \rightarrow B}$	Not derived; hypothetical mathematical object.
Energy accounting	Not yet demonstrated; $\Delta E_{\text{total}} = 0$ remains required.
Backward temporal transport	Assumed conditionally, not derived.
Experimental distinction	No established prediction separating BPTT from existing alternatives.
E-CTC comparison	Requires equation-level relative-state analysis.

8. Conservative Contribution Statement

BPTT is best understood at present as a **conditional phenomenological and mathematical framework** for source-history-preserving backward temporal displacement. Its central condition gives a precise statement of the no-retro-modification requirement, while the manuscript explicitly recognizes that related branching, CTC, decoherence, and alternate-history ideas already exist.

Key references: Deutsch (1991), Phys. Rev. D 44, 3197–3217; Shoshany & Stober (2023), arXiv:2303.07635; Hawking (1992), Phys. Rev. D 46, 603–611; Lewis (1976), "The Paradoxes of Time Travel."